

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES				
10							***** VALVES *****			810	NIPPLE TO BE CUT FROM SCH 160 PIPE.			
20							***** MODULAR VALVES *****			817	USE ONLY FOR TIE IN			
30										825	USE ONLY FOR MODELLING PURPOSE AT TIE-IN POINT.DIMENSIONAL DATA NOT CONFIRMED.			
30	U04		MODULAR DBB	K06AC02	0.5	1.5	MODULAR VALVE DB&B BALL TYPE 600# RF LTCS A350 LF2 CL.1 316 SS TRIM METAL LO - FULL SIZE TYPE 2BV	0	0	B1	ALL BOLTS ARE METRIC DIA.			
40	U04		MODULAR DBB	K06AC06	2	2	MODULAR VALVE DB&B BALL TYPE 600# RF LTCS A350 LF2 CL.1 316 SS TRIM METAL LO - FULL SIZE TYPE 2BV	0	0	B2	7/8" X 145MM BOLTS ONLY TO BE USED FOR 1 INCH 1500# PERAR BALL VALVES ITEM CODE B15AM10			
50	U04	VAV	MODULAR DBB	K06AC06	3	3	MODULAR VALVE DB&B BALL TYPE 600# RF LTCS A350 LF2 CL.1 316 SS TRIM METAL LO - FULL SIZE TYPE 2BV	0	0	UK9	DO NOT USE FOR ORIFICE FLANGE TAP ASSEMBLIES.			
60	U04		MODULAR DBB	M06AC02	0.5	1.5	MODULAR VALVE DB&B BALL TYPE 600# RF LTCS A350 LF2 CL.1 316 SS TRIM METAL LO - CONTROLLED BORE TYPE 2BVR	0	0	VA5	USE THESE VALVES ONLY WHERE SPECIFIED ON PIPING AND INSTRUMENT DIAGRAMS			
70	U04		MODULAR DBB	M06AC06	2	2	MODULAR VALVE DB&B BALL TYPE 600# RF LTCS A350 LF2 CL.1 316 SS TRIM METAL LO - CONTROLLED BORE TYPE 2BVR	0	0	VAV	USAGE IS LIMITED TO INSTRUMENT ISOLATION PURPOSE.			
80	U04	VAW	MODULAR DBB	M06EC02	0.75	1	MODULAR VALVE DB&B BALL TYPE 600# RF LTCS A350 LF2 CL.1 316 SS TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	2	2	VAW	VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055 - VENT DRAIN AND INSTRUMENT CONNECTION DETAILS.			
90	U04	VAW	MODULAR DBB	M06EC02	1	1	MODULAR VALVE DB&B BALL TYPE 600# RF LTCS A350 LF2 CL.1 316 SS TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	3	3	VB2	SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C.			
100							***** BALL VALVES *****							
100		VB2												
110														
110	U04		BALL	A06AC05	2	4	BALL 600# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION SOFT RB LO	0	0					
120	U04		BALL	A06AC06	2	4	BALL 600# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION METAL RB LO	0	0					
130	U04		BALL	A06AC07	6	6	BALL 600# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION SOFT RB GO	0	0					
140	U04		BALL	A06AC08	6	6	BALL 600# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION METAL RB GO	0	0					
150	U04		BALL	B06AC01	0.75	1.5	BALL 600# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING SOFT FB LO	0	0					
160	U04		BALL	B06AC02	0.75	1.5	BALL 600# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING METAL FB LO	0	0					
170	U04		BALL	B06AC05	2	4	BALL 600# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION SOFT FB LO	0	0					
180	U04	VA5	BALL	B06AC06	2	4	BALL 600# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION METAL FB LO	0	0					
190	U04	VA5	BALL	B06AC07	4	6	BALL 600# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION SOFT FB GO	0	0					
200	U04	VA5	BALL	B06AC07	24	24	BALL 600# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION SOFT FB GO	0	0					
210	U04	VA5	BALL	B06AC08	4	6	BALL 600# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION METAL FB GO	0	0					
220	U04	VA5	BALL	B06AC08	24	24	BALL 600# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION METAL FB GO	0	0					
230	U04	825, 817	BALL	BC5C3	1	1	BALL 600# RTJ LTCS A350 LF2 CL.1 316SS TRIM FS RB LONG PATT	0	0					
240							***** RISING STEM BALL VALVES *****							
240														
250														
250		VA5	BALL	R06AC06	0.75	6	BALL RISING STEM 600# RF LTCS A352 LCC 316SS TRIM METAL RB HO	0	0					
260							***** BUTTERFLY VALVES *****							
260														
270														
270	U04	VA5		T06AC61	6	6	BUTTERFLY 600# RF LTCS A352 LCC 316SS TRIM TOV METAL DBL FLG SHORT PAT GO	0	0					
280			BUTTERFLY	T06AC61	24	24	BUTTERFLY 600# RF LTCS A352 LCC 316SS TRIM TOV METAL DBL FLG SHORT PAT GO	0	0					
290							***** GATE VALVES *****							
290														
300														
300			GATE	W06AC70	3	4	GATE 600# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB HW	0	0					
310			GATE	W06AC71	6	6	GATE 600# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB GO	0	0					
320			GATE	W06AC71	24	24	GATE 600# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB GO	0	0					
330							***** GLOBE VALVES *****							
330														
340														
340	U04		GLOBE	L06AC48	0.75	1.5	GLOBE 600# RF LTCS A350 LF2 CL.1 TRIM 16 OS&Y HW	0	0					
350			GLOBE	L06AC41	2	4	GLOBE 600# RF LTCS A352 LCC TRIM 16 OS&Y HW	0	0					
360			GLOBE	L06AC42	6	6	GLOBE 600# RF LTCS A352 LCC TRIM 16 OS&Y GO	0	0					
370							***** CHECK VALVES *****							
370														
380														
380	U04		CHECK	C06AC50	0.75	1.5	CHECK PISTON 600# RF LTCS A350 LF2 CL.1 TRIM 16	0	0					
390			CHECK	C06AC45	2	6	CHECK DUAL WAFER LUG 600# RF LTCS A352 LCC TRIM 16 THROU BOLTED	0	0					
400			CHECK	C06AC52	24	24	CHECK DUAL DBL FLG 600# RF LTCS A352 LCC TRIM 16	0	0					
410							***** PIPE ABOVE GROUND *****							
410														
420														
420			PIPE	5358508	0.75	1.5	PIPE SCH XS SMLS LTCS A333 GR.6 PE	0	0					
430			PIPE	5358514	2	2	PIPE SCH XS SMLS LTCS A333 GR.6 BE	0	0					
440	U04	UK9	PIPE	5358511	0.5	0.75	PIPE SCH XXS SMLS LTCS A333 GR.6 PE	0	0					
450			PIPE	5358505	3	3	PIPE SCH STD SMLS LTCS A333 GR.6 BE	0	0					
460														
460			PIPE	9001257	4	4	LINEPIPE SMLS WT= 6.02 MM API-5L X60 BE	0	0					
470			PIPE	9001254	6	6	LINEPIPE SMLS WT= 6.35MM API-5L X60 BE	0	0					
480			PIPE	9001491	24	24	LINEPIPE SMLS WT = 14.27 MM API-5L X60 BE	0	0					
490			PIPE	9005244	28	28	LINEPIPE SMLS WT = 15.88 MM API-5L X60 BE	0	0					
			TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN		CONSTRUCTION		CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
					ALL SIZES - BUTT WELD PWHT - PIPE SIZE 24" AND LARGER SEAL WELDING - REQUIRED PIPE BENDS - INDUCTION BENDS THREAD SEALANT - PTFE PASTE		CORR. ALLOW: 1.5 MM	DESIGN: ASME B16.5 CODE: ASME B31.8 MAX. DESIGN PARAMETERS (LIMITED BY PIPE WT) - 93 BARG @ 80 DEG. C - 75 BARG. @ 100 DEG.C FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS FLANGE RATING: 600 RF TEMPERATURE RANGE: -40 / 100 °C COMMENT: 93 BARG @ 80 DEGREES C	12/16/2024	BY OZAC	U04	Page 1 of 4	600H5P

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500										
500		810	PIPE	5358512	0.5	0.5	PIPE SCH 160 SMLS LTCS A333 GR.6 PE	0	0	
510										
510	U04		PIPE	9001488	12	12	LINEPIPE SMLS WT = 11.13 MM API-5L X60 BE	0	0	
520	U04		PIPE	9001488	18	18	LINEPIPE SMLS WT = 11.13 MM API-5L X60 BE	0	0	
530	U04		PIPE	9001487	10	10	LINEPIPE SMLS WT = 8.74 MM API-5L X60 BE	0	0	
540	U04		PIPE	9001498	8	8	LINEPIPE SMLS WT= 8.38 MM API-5L X60 BE	0	0	
550										
550							***** PIPE BELOW GROUND *****			
560										
560			PIPE	9001257CTD	4	4	LINEPIPE SMLS WT= 6.02 MM API-5L X60 BE COATED	0	0	
570			PIPE	9001254CTD	6	6	LINEPIPE SMLS WT= 6.35 MM API-5L X60 BE COATED	0	0	
580			PIPE	9001491CTD	24	24	LINEPIPE SMLS WT = 14.27 MM API-5L X60 BE COATED	0	0	
590	U04		PIPE	9001488CTD	12	12	LINEPIPE SMLS WT = 11.13 MM API-5L X60 BE COATED	0	0	
600	U04		PIPE	9001493CTD	8	8	LINEPIPE SMLS WT = 9.53 MM API-5L X60 BE COATED	0	0	
610	U04		PIPE	9001498CTD	8	8	LINEPIPE SMLS WT= 8.38 MM API-5L X60 BE COATED	0	0	
620										
620							***** FITTINGS ABOVE & BELOW GROUND *****			
630										
630			ELBOW	5392600	0.75	2	ELL BW 90 DEG LR SCH XS LTCS A420 WPL6	0	0	
640	U04	UK9	ELBOW	5392815	0.5	0.75	ELL BW 90 DEG LR SCH XXS LTCS A420 WPL6	0	0	
650			ELBOW	5392438	3	3	ELL BW 90 DEG LR SCH STD LTCS A420 WPL6	0	0	
660			ELBOW	9006566	4	4	ELL BW 90 DEG LR WT 6.02 MM A860 WPHY 60	0	0	
670			ELBOW	9002190	6	6	ELL BW 90 DEG LR WT 6.35 MM A860 WPHY 60	0	0	
680	U04		ELBOW	9006468	12	12	ELL BW 90 DEG LR WT 11.13 MM A860 WPHY 60	0	0	
690	U04		ELBOW	9002157	10	10	ELL BW 90 DEG LR WT 8.74MM A860 WPHY 60	0	0	
700	U04		ELBOW	9006471	8	8	ELL BW 90 DEG LR WT 9.53 MM A860 WPHY 60	0	0	
710			ELBOW	5392606	0.75	2	ELL BW 45 DEG LR SCH XS LTCS A420 WPL6	0	0	
720	U04	UK9	ELBOW	5392831	0.5	0.75	ELL BW 45 DEG LR SCH XXS LTCS A420 WPL6	0	0	
730			ELBOW	5392444	3	3	ELL BW 45 DEG LR SCH STD LTCS A420 WPL6	0	0	
740			ELBOW	9006573	4	4	ELL BW 45 DEG LR WT 6.02MM A860 WPHY 60	0	0	
750	U04		ELBOW	9006571	6	6	ELL BW 45 DEG LR WT 6.35MM A860 WPHY 60	0	0	
760	U04		ELBOW	9006593	10	10	ELL BW 45 DEG LR WT 8.74MM A860 WPHY 60	0	0	
770										
770			TEE	5392609	0.75	2	TEE BW SCH XS LTCS A420 WPL6	0	0	
780			TEE	5392447	3	3	TEE BW SCH STD LTCS A420 WPL6	0	0	
790	U04	UK9	TEE	5392879	0.5	0.75	TEE SCH XXS LTCS A420 WPL6	0	0	
800			TEE	9301610	4	4	TEE BW WT 6.02 MM LTCS A860 WPHY-60	0	0	
810			TEE	9301609	6	6	TEE BW WT 6.35 MM LTCS A860 WPHY-60	0	0	
820	U04		TEE	9006487	10	10	TEE BW WT 8.74MM A860 WPHY 60-W	0	0	
830			TEE RED	5392661	0.75	1.5	TEE RED BW SCH XS LTCS A420 WPL6	1	2	
840			TEE RED	5392454	1	2	TEE RED BW SCH STD/XS LTCS A420 WPL6	3	3	
850			TEE RED	9006664	2	2	TEE RED BW WT 6.02 MM / SCH XS LTCS A860 WPHY 60	4	4	
860			TEE RED	9006659	3	3	TEE RED BW WT 6.02 MM / STD LTCS A860 WPHY 60	4	4	
870			TEE RED	9006658	3	3	TEE RED BW WT 6.35 MM / STD LTCS A860 WPHY 60	6	6	
880			TEE RED	9006605	4	4	TEE RED BW WT 6.35 MM / 6.02 MM LTCS A860 WPHY 60	6	6	
890	U04		TEE RED	9004123	6	6	TEE RED BW WT 8.74 MM/6.35 MM LTCS A860 WPHY 60	10	10	
900										
900			CAP	5392610	0.75	2	CAP BW SCH XS LTCS A420 WPL6	0	0	
910			CAP	5392448	3	3	CAP BW SCH STD LTCS A420 WPL6	0	0	
920			CAP	9005062	4	4	CAP BW WT 6.02 MM LTCS A860 WPHY-60	0	0	
930			CAP	9005055	6	6	CAP BW WT 6.35MM LTCS A860 WPHY-60	0	0	
940	U04		CAP	9005064	12	12	CAP BW WT 11.13 MM LTCS A860 WPHY-60	0	0	
950	U04		CAP	9005059	10	10	CAP BW WT 8.74 MM LTCS A860 WPHY-60	0	0	
960			CAP	9005068	24	24	CAP BW WT 14.27 MM LTCS A860 WPHY-60	0	0	
970			CAP	9005069	28	28	CAP BW WT 15.88 MM LTCS A860 WPHY-60	0	0	
980										
980			RED CONC	5392458	0.75	1.5	REDUCER CONC SCH XS LTCS A420 WPL6	1	2	
990			RED CONC	5392452	1.5	2	REDUCER CONC SCH STD/XS LTCS A420 WPL6	3	3	
1000			RED CONC	9006918	1.5	3	REDUCER CONC WT 6.02 MM / XS WT LTCS A860 WPHY-60	4	4	
1010			RED CONC	9006915	3	3	REDUCER CONC WT 6.35 MM / XS LTCS A860 WPHY-60	6	6	
1020	U04			9002199	10	10	REDUCER CONC WT 11.1MM / 8.7MM A860 WPHY 60	12	12	
1030	U04		RED CONC	9006929	6	6	REDUCER CONC WT 8.74 MM / 6.35MM LTCS A860 WPHY-60	10	10	
1040			RED ECC	5392461	0.75	1.5	REDUCER ECC SCH XS LTCS A420 WPL6	1	2	
1050			RED ECC	5392453	1.5	2	REDUCER ECC SCH STD/XS LTCS A420 WPL6	3	3	
1060			RED ECC	9001372	1.5	3	REDUCER ECC WT 6.02 MM / XS WT LTCS A860 WPHY-60	4	4	
1070	U04		RED ECC	9002510	3	3	REDUCER ECC WT 6.35 MM/STD LTCS A860 WPHY-60	6	6	
1080			RED ECC	9002312	4	4	REDUCER ECC WT 6.35 MM / 6.02 MM LTCS A860 WPHY-60	6	6	
1090			RED ECC	9001386	24	24	REDUCER ECC WT 15.88 MM / 14.27MM LTCS A860 WPHY-60-W	28	28	
1100	U04			9002229	10	10	REDUCER ECC WT 11.1MM / WT 8.7MM LTCS A860 WPHY-60	12	12	
1110	U04		RED ECC	9004131	6	6	REDUCER ECC WT 8.74 MM / 6.35 LTCS A860 WPHY-60-W	10	10	
1120							***** INDUCTION BENDS ABOVE GROUND *****			

	TENGIZCHEVROI TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD PWHT - PIPE SIZE 24" AND LARGER SEAL WELDING - REQUIRED PIPE BENDS - INDUCTION BENDS THREAD SEALANT - PTFE PASTE	CORR. ALLOW: 1.5 MM	DESIGN: ASME B16.5 CODE: ASME B31.8 MAX. DESIGN PARAMETERS (LIMITED BY PIPE WT) - 93 BARG @ 80 DEG. C - 75 BARG. @ 100 DEG.C FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS FLANGE RATING: 600 RF TEMPERATURE RANGE: -40 / 100 °C COMMENT: 93 BARG @ 80 DEGREES C	12/16/2024	BY OZAC	U04	Page 2 of 4	600H5P

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES
1130										
1130			INDUC BEND	9001810	4	4	INDUCTION BEND 90 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
1140			INDUC BEND	9001813	6	6	INDUCTION BEND 90 DEG X 5D RADIUS WT 6.35MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
1150			INDUC BEND	9005322	24	24	INDUCTION BEND 90 DEG X 5D RADIUS WT 14.27 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
1160	U04		INDUC BEND	9007038CTD	12	12	INDUCTION BEND 90 DEG X 10D RADIUS WT 11.13MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1170	U04		INDUC BEND	9007031CTD	6	6	INDUCTION BEND 90 DEG X 10D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1180	U04		INDUC BEND	9007043CTD	6	6	INDUCTION BEND 90 DEG X 15D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1190	U04		INDUC BEND	9007050CTD	6	6	INDUCTION BEND 90 DEG X 20D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1200			INDUC BEND	9001820	4	4	INDUCTION BEND 45 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
1210			INDUC BEND	9001823	6	6	INDUCTION BEND 45 DEG X 5D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
1220			INDUC BEND	9005328	24	24	INDUCTION BEND 45 DEG X 5D RADIUS WT 14.27 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
1230			INDUC BEND	9001830	4	4	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
1240			INDUC BEND	9001833	6	6	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
1250			INDUC BEND	9005334	24	24	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 14.27 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
1260										
1260							***** INDUCTION BENDS BELOW GROUND *****			
1270			INDUC BEND	9001810CTD	4	4	INDUCTION BEND 90 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1280			INDUC BEND	9001813CTD	6	6	INDUCTION BEND 90 DEG X 5D RADIUS WT 6.35MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1290			INDUC BEND	9005322CTD	24	24	INDUCTION BEND 90 DEG X 5D RADIUS WT 14.27 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1300			INDUC BEND	9001820CTD	4	4	INDUCTION BEND 45 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1310			INDUC BEND	9001823CTD	6	6	INDUCTION BEND 45 DEG X 5D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1320			INDUC BEND	9005328CTD	24	24	INDUCTION BEND 45 DEG X 5D RADIUS WT 14.27 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1330	U04		INDUC BEND	9007010CTD	24	24	INDUCTION BEND 45DEG X 10D RADIUS WT 14.27 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1340	U04		INDUC BEND	9007001CTD	6	6	INDUCTION BEND 45DEG X 10D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1350	U04		INDUC BEND	9007025CTD	24	24	INDUCTION BEND 45DEG X 15D RADIUS WT 14.27 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1360	U04		INDUC BEND	9007016CTD	6	6	INDUCTION BEND 45DEG X 15D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1370			INDUC BEND	9001830CTD	4	4	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1380			INDUC BEND	9001833CTD	6	6	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1390			INDUC BEND	9005334CTD	24	24	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 14.27 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1400										
1400							***** OLETS ABOVE & BELOW GROUND *****			
1410										
1410	U04		OLETS	9006219	0.75	1.5	WELDOLET WT 6.02MM/SCH XS LTCS A694 F60	4	4	
1420	U04		OLETS	5505179	0.75	1	WELDOLET SCH STD/XS LTCS A350 LF2 CL.1	3	3	
1430	U04		OLETS	9002471	1	1	WELDOLET WT 14.27 MM/ SCH XS WT LTCS A694 F60	24	24	
1440	U04		OLETS	9006422	3	3	WELDOLET WT 8.74 MM/STD WT LTCS A694 F60	10	10	
1450	U04		OLETS	9006423	2	2	WELDOLET WT 8.74 MM/XS WT LTCS A694 F60	10	10	
1460			OLETS	9006407	0.75	2	WELDOLET WT 6.35 MM/XS WT LTCS A694 F60	6	6	
1470			OLETS	9006820	3	3	WELDOLET WT 19.05 MM/STD WT LTCS A694 F60	24	24	
1480			OLETS	9005044	4	4	WELDOLET WT 14.27 MM/ 6.02 MM LTCS A694 F60	24	24	
1490			OLETS	9002473	6	6	WELDOLET WT 14.27 MM/ 6.35 MM LTCS A694 F60	24	24	
1500			OLETS	9002819	8	8	WELDOLET WT 14.27 MM/ 7.92 MM LTCS A694 F60	24	24	
1510			OLETS	9005046	4	4	WELDOLET WT 15.88 MM/ 6.02 MM LTCS A694 F60	28	28	
1520			OLETS	9002088	6	6	WELDOLET WT 15.88 MM/ 6.35 MM LTCS A694 F60	28	28	
1530			OLETS	9005047	8	8	WELDOLET WT 15.88 MM/ 7.92 MM LTCS A694 F60	28	28	
1540			OLETS	9002086	10	10	WELDOLET WT 15.88 MM/ 8.74 MM LTCS A694 F60	28	28	
1550										
1550	U04		OLETS	5504006	0.75	1.5	NIPOFLANGE 600# RF SCH XS LTCS A350 LF2 CL.1	2	2	
1560	U04		OLETS	5504007	0.75	2	NIPOFLANGE 600# RF SCH STD/XS LTCS A350 LF2 CL.1	3	3	
1570			OLETS	9005030	0.75	2	NIPOFLANGE 600# RF 6.02 MM/XS WT LTCS A694 F60	4	4	
1580			OLETS	9004270	0.75	2	NIPOFLANGE 600# RF 6.35 MM/SCH XS WT LTCS A694 F60	6	6	
1590			OLETS	9005105	0.75	2	NIPOFLANGE 600# RF 14.27 MM/XS WT LTCS A694 F60	24	24	
1600			OLETS	9005034	0.75	2	NIPOFLANGE 600# RF 15.88 MM/XS WT LTCS A694 F60	28	28	
1610	U04		OLETS	9005032	0.75	1.5	NIPOFLANGE 600# RF 8.74 MM/XS WT LTCS A694 F60	10	10	
1620	U04		OLETS	9003460	1	1	NIPOFLANGE 600# RF WT 11.13 MM/SCH XS WT LTCS A694 F60	12	12	
1630			OLETS	9006821	2	2	NIPOFLANGE 1500# RF WT 6.35 MM/ SCH 160 LTCS A694 F60	6	6	
1640										
1640							***** FLANGES ABOVE & BELOW GROUND*****			
1650										
1650	U04		FLG ADAPTER	5851204	0.5	2	FLANGE ADAPTER 600# RF 316 SS X 12 MM TUBE	0	0	
1660										

	TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD PWHT - PIPE SIZE 24" AND LARGER SEAL WELDING - REQUIRED PIPE BENDS - INDUCTION BENDS THREAD SEALANT - PTFE PASTE	CORR. ALLOW: 1.5 MM	DESIGN: ASME B16.5 CODE: ASME B31.8 MAX. DESIGN PARAMETERS (LIMITED BY PIPE WT) - 93 BARG @ 80 DEG. C - 75 BARG. @ 100 DEG.C FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS FLANGE RATING: 600 RF TEMPERATURE RANGE: -40 / 100 °C COMMENT: 93 BARG @ 80 DEGREES C	12/16/2024	BY OZAC	U04	Page 3 of 4	600H5P

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES
1660	U04	UK9	FLANGE	5556976	0.5	0.75	FLG WN 600# RF LTCS A350 LF2 CL.1 SCH XXS	0	0	
1670	U04		FLANGE	5557036	0.5	2	FLG WN 600# RF LTCS A350 LF2 CL.1 SCH XS	0	0	
1680	U04		FLANGE	5557033	3	3	FLG WN 600# RF LTCS A350 LF2 CL.1 SCH STD	0	0	
1690			FLANGE	9001723	4	4	FLG WN 600# RF LTCS A694 F60 WT 6.02MM	0	0	
1700			FLANGE	9001471	6	6	FLG WN 600# RF LTCS A694 F60 WT 6.35MM	0	0	
1710			FLANGE	9001515	24	24	FLG WN 600# RF LTCS A694 F60 WT 14.27MM	0	0	
1720	U04		FLANGE	9003487	18	18	FLG WN 600# RF LTCS A694 F60 WT 11.13 MM	0	0	
1730	U04		FLANGE	9003487	12	12	FLG WN 600# RF LTCS A694 F60 WT 11.13 MM	0	0	
1740	U04		FLANGE	9003484	10	10	FLG WN 600# RF LTCS A694 F60 WT 8.74 MM	0	0	
1750	U04		FLANGE	9003485	8	8	FLG WN 600# RF LTCS A694 F60 WT 9.53 MM	0	0	
1760	U04		BLIND	5615598	0.5	3	FLG BLIND 600# RF LTCS A350 LF2 CL.1	0	0	
1770			BLIND	9001518	4	24	FLG BLIND 600# RF LTCS A694 F60	0	0	
1780	U04		TEE RED	5599705	0.5	0.5	FLG SCRD (FNPT) REDUCING 600# RF A350 LF2 CL.1	0.75	1.5	
1790	U04	817	FLANGE	5558036	1.5	3	FLG WN 600# RTJ LTCS A350 LF2 CL.1 SCH XS	0	0	
1800	U04	817	FLANGE	9006724	4	4	FLG WN 600# RTJ A694 F60 WT 6.02MM	0	0	
1810	U04	817	FLANGE	9006704	6	6	FLG WN 600# RTJ A694 F60 WT 6.35MM	0	0	
1820	U04	817	BLIND	5615698	0.5	3	FLG BLIND 600# RTJ LTCS A350 LF2 CL.1	0	0	
1830	U04		FLANGE	5557393	1	2	FLG WN 1500# RF LTCS A350 LF2 CL.1 SCH XS	0	0	
1840			FLANGE	9005389	4	4	FLG WN 900# RF LTCS A694 F60 WT 6.02MM	0	0	
1850			FLANGE	9006755	6	6	FLG WN 900# RF LTCS A694 F60 WT 6.35MM	0	0	
1860	U04		FLANGE	5555935	3	3	FLG WN 900# RF LTCS A350 LF2 CL.1 SCH STD	0	0	
1870										
1870			ORIFICE FLG	9001556	6	6	WN ORIF SET 600# RF A694 F60 WT 6.35MM	0	0	
1880	U04		ORIFICE FLG	9001557	4	4	WN ORIF SET 600# RF A694 F60 WT 6.02MM	0	0	
1890	U04		ORIFICE FLG	9001566	10	10	WN ORIF SET 600# RF A694 F60 WT 8.74MM	0	0	
1900										
1900						***** GASKETS*****				
1910										
1910			GASKET	5673103	0.5	24	GASKET 600# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
1920	U04		GASKET	5672251	1	2	GASKET 1500# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
1930	U04		GASKET	5673117	3	6	GASKET 900# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
1940	U04	817	GASKET	5673202	1	6	GASKET 600# RTJ SOFT IRON TEFLON COATED OCTAGONAL RING	0	0	
1950										
1950		B1				*****BOLTING*****				
1960										
1960	U04		BOLTING	5675770	0.5	24	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS	0	0	
1970										
1970			BOLTING	5675675	0.75	24	JACK SCREW HEX HEAD A320 GR L7M	0	0	
1980										
1980						***** BLINDS & SPACERS *****				
1990										
1990	U04		SPEC BLIND	5749587	0.75	8	SPECTACLE BLIND 600# RF LTCS A516-GR70	0	0	
2000										
2000	U04		PADDLE	5750040	12	24	PADDLE BLANK 600# RF LTCS A516-GR70	0	0	
2010	U04		PADDLE	5750251	12	24	PADDLE SPACER 600# RF LTCS A516-GR70	0	0	
2020										
2030										
2030	U04	817	FLANGE	SP-2648	2	2	FLANGE TRANSITION 600H5P/10000H5A COMPACT FLG RJ A350 LF2 CL.1 SCH XS	0	0	

	TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
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